

WolfTracker System Test Plan

Document Author: Jayani Sivakumar and Diya Patel

Date: 11-November-2024

Introduction

This system test plan verifies the core functionalities of the **WolfTracker** project, which allows users to log and track project-related tasks by category. The key functionalities include creating and loading projects, saving project data, and logging tasks, along with displaying their statistics. The test plan will guide users through operations including creating, modifying, and viewing task logs in the application's GUI. To run the tests:

1. Right click on WolfManagerGUI class in the Package Explorer.
2. Select Run As > Java Application

Test Files

The file below with the following contents will be used for testing:

project1.txt

```
! CSC 216 Task Log

# Debugging

# Deployment

# Design

# Documentation

# Implementation

# System Test

# Unit Test

* Read Project 2 requirements,45,Design

Read Project 2 requirements
```

(<https://pages.github.ncsu.edu/engr-csc216-staff/CSC216-SE-Materials/projects/project2/project2-part1.html>)

and identify candidate classes and methods.

* Created CRC Cards,27,Design

Identified the key classes and created CRC cards. Noted responsibilities, collaborators, and possible state.

* Transferred CRC Cards to UMLetino,35,Design

Started creating a UML class diagram from the requirements

* Downloaded design proposal and rational template,5,Design

See (<https://pages.github.ncsu.edu/engr-csc216/projects/project2/project2-part1.html>) for template link

* Wrote design proposal and rationale,75,Design

Started with UML class diagram description. Incorporated feedback from Project 1.

* Identify 5 system tests,20,System Test

Consider 5 major paths through the system when working with projects, logs, and tasks. How would I use the system to keep track of my tasks?

* Created P2P2 skeleton,48,Implementation

Created the skeleton for P2P2.

* Added Javadoc,65,Documentation

Javadoced skeleton

* Implemented & Tested POJO,30,Implementation,Unit Test

Implemented the POJO class. Added unit tests to make sure everything is correct.

* Implemented file reader class,35,Implementation

Started implementing the reader class by reading in all the input file and breaking into top-level tokens.

* Tested reader class,32,Unit Test

Found a bug in my implementation when breaking the text into tokens.

* Debugging Reader Bug,15,Debugging

Debugged my file reader bug.

* Ran System Tests,20,System Test

Ran system tests on project and recorded actual results. They all passed!

* Generated Javadoc,15,Documentation

Generated Javadoc and pushed to GitHub finishing the project!

System Tests

Test ID	Description	Expected Results	Actual Results
Test 1: Load a Valid Test File and provide Invalid Input for Task Logging (UC12) Tests Boundary class Author: Jayani Sivakumar And Diya Patel	Preconditions: None 1. Run WolfManager GUI 2. Select the option from the File menu to load from a file. 3. From the File chooser, browse and select for project1.txt 4. Select the category "Implementation" option from the current category dropdown to	Step 5 Results: WolfManager GUI loads. Project Name is displayed as follows: CSC 216 Task Log Summary statistics is displayed as follows: Number of logged tasks: 14 Min Duration: 5 Max Duration: 75 Average Duration: 33.4	Output: Step 5 Results: The WolfManager GUI loads. After running the WolfManager GUI, it successfully loads the mentioned file in Step 3 after clicking on the "Load" button and choosing the 'project1.txt' file from the test-files. Initially the list of

	choose the category. 5. <i>Check results</i> 6. Select the task to edit under the Logged Tasks section (Created P2P2 skeleton,48,Implementation) 7. Under the task section, edit the task Duration to an invalid one (-20). 8. Click on the “Add/Edit” button at the end of the task section. 9. <i>Check results</i> 10. Close GUI	Initially the list of logged tasks are displayed in the menu as follows: Read Project 2 requirements,45,Design Created CRC Cards,27,Design Transferred CRC Cards to UMLetino,35,Design Downloaded design proposal and rational template,5,Design Wrote design proposal and rationale,75,Design Identify 5 system tests,20,System Test Created P2P2 skeleton,48,Implementation Added Javadoc,65,Documentation Implemented & Tested POJO,30,Implementation,Unit Test Implemented file reader class,35,Implementation Tested reader class,32,Unit Test	logged tasks are displayed in the menu as follows: Read Project 2 requirements,45,Design Created CRC Cards,27,Design Transferred CRC Cards to UMLetino,35,Design Downloaded design proposal and rational template,5,Design Wrote design proposal and rationale,75,Design Identify 5 system tests,20,System Test Created P2P2 skeleton,48,Implementation Added Javadoc,65,Documentation Implemented & Tested POJO,30,Implementation,Unit Test Implemented file reader class,35,Implementation Tested reader class,32,Unit Test
--	--	---	---

		Debugging Reader Bug,15,Debugging Ran System Tests,20,System Test Generated Javadoc,15,Documentation Step 9 Results: After selecting the "Add/Edit" button, we get an error message displaying "Incomplete task log information." No change is reflected in Logged Tasks/Summary Statistics sections and task edit area is cleared.	Debugging Reader Bug,15,Debugging Ran System Tests,20,System Test Generated Javadoc,15,Documentation Step 9 Results: After selecting the "Add/Edit" button, we get an error message displaying "Incomplete task log information." as the duration is an invalid one. No change is reflected in Logged Tasks/Summary Statistics sections and task edit area is cleared.
Test 2: Edit an Existing Task (UC13) Tests Equivalence class Author: Jayani Sivakumar and Diya Patel	Preconditions: Test 1 has passed. 1. Run WolfManager GUI 2. Select the category "Design" option from the current category dropdown to choose the category. 3. Select the task to	Step 6 results: WolfManager GUI loads. Summary statistics is updated and displayed as follows: Number of logged tasks: 5 Min Duration: 5 Max Duration: 75 Average Duration: 41.0	Output: Step 6 results: WolfManager GUI loads. Summary statistics is updated and displayed as follows: Number of logged tasks: 5 Min Duration: 5 Max Duration: 75

	edit under the Logged Tasks section (Created CRC Cards,27,Design) 4. Under the task section, change the task Duration to 45, and Description to "Updated CRC cards based on new feedback." 5. Click on the "Add/Edit" button at the end of the task section. 6. <i>Check results</i> 7. Close GUI	The list of logged tasks are updated and displayed in the menu as follows: Read Project 2 requirements,45, Design Created CRC Cards,45,Design Transferred CRC Cards to UMLetino,35,Design Downloaded design proposal and rational template,5,Design Wrote design proposal and rationale,75,Design Task edit area is shown as follows: Task Title: Created CRC Cards Duration: 45 Description: Updated CRC cards based on new feedback.	Average Duration: 41.0 The list of logged tasks are updated and displayed in the menu as follows: Read Project 2 requirements,45, Design Created CRC Cards,45,Design Transferred CRC Cards to UMLetino,35,Design Downloaded design proposal and rational template,5,Design Wrote design proposal and rationale,75,Design Task edit area is shown as follows: Task Title: Created CRC Cards Duration: 45 Description: Updated CRC cards based on new feedback.
--	--	---	---

Test 3: Add New Category with Duplicate Name (UC8) Tests Boundary class Author: Jayani Sivakumar and Diya Patel	Preconditions: Test 1 has passed 1. Run WolfManager GUI 2. Under the category section, select the “Add Category” button. 3. Enter “Debugging” and click on the “OK” button. 4. <i>Check results</i> 5. Close GUI	Step 4 Results: WolfManager GUI loads. Error message displayed saying “Invalid name.” and the user is returned to prompt to change the name. Current category dropdown is not updated and displayed as follows: All Tasks Debugging Design Documentation Implementation System Test Unit Test Logged tasks are not updated and displayed in the menu as follows: Read Project 2 requirements,45, Design Created CRC Cards,27,Design Transferred CRC Cards to UMLetino,35,Design Downloaded design proposal and rational template,5,Design	Output: Step 4 Results: WolfManager GUI loads. Error message displayed saying “Invalid name.” and the user is returned to prompt to change the name as we are trying to add a duplicate category. Current category dropdown is not updated and displayed as follows: All Tasks Debugging Design Documentation Implementation System Test Unit Test Logged tasks are not updated and displayed in the menu as follows: Read Project 2 requirements,45, Design Created CRC Cards,27,Design Transferred CRC Cards to UMLetino,35,Design
--	--	---	--

		Wrote design proposal and rationale,75,Design Identify 5 system tests,20,System Test Created P2P2 skeleton,48,Implementation Added Javadoc,65,Documentation Implemented & Tested POJO,30,Implementation,Unit Test Implemented file reader class,35,Implementation Tested reader class,32,Unit Test Debugging Reader Bug,15,Debugging Ran System Tests,20,System Test Generated Javadoc,15,Documentation	Downloaded design proposal and rational template,5,Design Wrote design proposal and rationale,75,Design Identify 5 system tests,20,System Test Created P2P2 skeleton,48,Implementation Added Javadoc,65,Documentation Implemented & Tested POJO,30,Implementation,Unit Test Implemented file reader class,35,Implementation Tested reader class,32,Unit Test Debugging Reader Bug,15,Debugging Ran System Tests,20,System Test Generated Javadoc,15,Documentation
--	--	---	--

Test 4: Attempt to Edit All Tasks Category (UC9) Tests Boundary class Author: Jayani Sivakumar And Diya Patel	Preconditions: Test 1 has passed. 1. Run WolfManager GUI 2. Under the category section, select the category “All Tasks” category. 3. Click on the “Edit Category” button. 4. <i>Check results</i> 5. Enter a new name in the prompt (“Jenkins”). 6. Click on the “OK” button. 7. <i>Check results</i> 8. Close GUI	Step 4 Results: WolfManager GUI loads. Error message is displayed saying “The All Tasks category may not be edited.” The All Tasks category remains unchanged. Current category dropdown is not updated and displayed as follows: All Tasks Debugging Design Documentation Implementation Jenkins System Test Unit Test Logged tasks are not changed and displayed in the menu as follows: Read Project 2 requirements,45, Design Created CRC Cards,27,Design Transferred CRC Cards to UMLetino,35,Desi gn Downloaded design proposal and rational	Output: Step 4 Results: WolfManager GUI loads. Error message is displayed saying “The All Tasks category may not be edited.” The All Tasks category remains unchanged as we cannot edit the All Tasks Category. Current category dropdown is not updated and displayed as follows: All Tasks Debugging Design Documentation Implementation Jenkins System Test Unit Test Logged tasks are not changed and displayed in the menu as follows: Read Project 2 requirements,45, Design Created CRC Cards,27,Design Transferred CRC Cards to
--	--	--	---

		template,5,Design Wrote design proposal and rationale,75,Design Identify 5 system tests,20,System Test Created P2P2 skeleton,48,Implementation Added Javadoc,65,Documentation Implemented & Tested POJO,30,Implementation,Unit Test Implemented file reader class,35,Implementation Tested reader class,32,Unit Test Debugging Reader Bug,15,Debugging Ran System Tests,20,System Test Generated Javadoc,15,Documentation Step 7 Results: Logged Tasks/SummaryStatistics sections and task edit area is empty.	UMLetino,35,Design Downloaded design proposal and rational template,5,Design Wrote design proposal and rationale,75,Design Identify 5 system tests,20,System Test Created P2P2 skeleton,48,Implementation Added Javadoc,65,Documentation Implemented & Tested POJO,30,Implementation,Unit Test Implemented file reader class,35,Implementation Tested reader class,32,Unit Test Debugging Reader Bug,15,Debugging Ran System Tests,20,System Test Generated Javadoc,15,Documentation Step 7 Results: Logged Tasks/SummarySt
--	--	---	---

			statistics sections and task edit area is empty as there are no tasks associated with "Jenkins" category yet.
Test 5: Create New Project with Existing Project Open (UC1) Tests Equivalence class Author: Jayani Sivakumar And Diya Patel	Preconditions: Test 1 has passed. 1. Run WolfManager GUI 2. In the main menu, select the option to Create a New Project. 3. Select "No" to the prompt asking for confirmation we want to create a new project to save the existing project and lose all existing changes. 4. Name the new project "Wolf Tracker Goat Log" and click on the OK button. 5. <i>Check results</i> 6. Close GUI	Step 5 Results: WolfManager GUI loads. The contents of the previous project is lost and the following is displayed: Project Name: Wolf Tracker Goat Log SummaryStatistics, Logged Tasks and Task sections are empty.	Output: Step 5 Results: WolfManager GUI loads. The contents of the previous project is lost and the following is displayed: Project Name: Wolf Tracker Goat Log SummaryStatistics, Logged Tasks and Task sections are empty.